

HARIKRISHNAN A

Data Scientist/Senior Data Analyst

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Passport Details : V5015476 (Valid till: 06/12/2031)

SUMMARY

Results-driven Data Scientist and Machine Learning Engineer with hands-on experience in leveraging cutting-edge AI models, including GPT models through Azure OpenAI, to build intelligent automation solutions. Proficient in applying NLP techniques such as BERT for predictive modeling, data extraction, and document classification. Adept at developing end-to-end solutions for tasks like automated data entry, anti-fraud detection, and travel recommendations using image recognition models. Experienced in Python, SQL, and cloud-based ML platforms for seamless model deployment and optimization.

EXPERIENCE

Data Science Analyst

Trane Technologies

01/2025 - Present Bangalore, India

HVAC and Refrigeration

- Developed and implemented **risk prediction models** achieving an accuracy rate of over 80% and leveraged accurate risk predictions to inform strategic decisions, resulting in increased business value.
- Experienced in designing, building, and deploying robust data pipelines using Dataiku for various data processing needs.

Senior Engineer - Data Science

Air India

05/2024 - 12/2024 Ernakulam

Aviation

- Experienced Machine Learning specializing in the development of Convolutional Neural Network (CNN) models to leverage data and extract valuable insights.
- Proficient in utilizing Azure Databricks for efficient data processing, and model training
- Dedicated to applying advanced machine learning techniques to solve complex problems and drive data-driven decision-making.
- Developed an automated **VISA scanner** solution using **Azure OpenAI (GPT-4o-mini)** to extract user details and auto-fill forms, reducing manual efforts.

Analyst - Business analytics

Allianz

07/2021 - 05/2024 Trivandrum

Insurance/Financial Service

- Data Analysis and Machine Learning Expertise: Strong foundation in Logistic and Linear Regression, SQL data manipulation, and Python programming.
- Deep Learning Proficiency: Specialized in implementing Convolutional Neural Networks (CNN) for image analysis and pattern recognition.
- Problem-Solving and Insight Generation: Leveraging advanced machine learning techniques to solve complex challenges and deliver data-driven solutions that support informed decision-making.
- Conducted regular reviews of operations and identified areas for improvement.
- Machine Learning Model Development: Created ML models from scratch for predictive modelling to classify fraud and non fraud cases.

Data Science Intern

Allianz

01/2021 - 07/2021 Trivandrum

EDUCATION

MSc Data Science

University Of Kerala

07/2019 - 05/2021

B.Sc. Mathematics

CMS College Kottayam

06/2016 - 05/2019

PROJECTS

Anti-Fraud Predictive Modeling

Employed **NLP techniques** (BERT model) to extract relevant features from unstructured text descriptions, enhancing the accuracy of anti-fraud detection. Designed and implemented a highly accurate predictive model with 98% accuracy using Python, Azure, Machine Learning, and SQL, effectively identifying potential fraud before it occurs.

Use case Classification & Travel Recommendation System

Developed a **document classification** and **travel recommendation system** using OpenAI that automatically detects and categorizes document types (e.g., passports, visas) from image uploads, utilizing CNNs and NLP for precise classification and enhanced accuracy.

Invoice Tampering Detection

Engineered a deep learning-based tool from scratch using Python and CNNs to detect tampered invoices and documents, significantly enhancing the accuracy of document verification processes.

Flight Turnaround Time Analytics

Developed an object detection model using Python, Streamlit, and YOLOv8 to track and analyze flight turnaround times, pinpointing delays by monitoring and timing various ground operations.

Queue Time Analysis

Created a CNN-based model to analyze and optimize queue times using real-time object detection on Raspberry Pi devices, significantly improving management efficiency and operational throughput.

SKILLS

Data Science Toolkits/Packages

Scikit, NumPy, Pandas, Matplotlib, Seaborn, SciPy

Programming Tools/Languages

Python, SQL

Data Visualization:

Tableau, MS Excel, PowerBI

Statistics and Machine Learning

Classification, Regression, Decision Tree, Random Forest, KNN, K-means

Deep Learning

CNN, YOLO, Detectron2, ANN

Data Analytics, Machine Learning, Predictive Modeling, Fraud Analytics, Problem Solving, Quantexa, Azure, Generative AI (Gen AI), Modeling Techniques, Docker, Dataiku

KEY ACHIEVEMENTS

 **Best Employee Award at Allianz**

 **Best All-Rounder Award at Allianz**

 **Gold Medal at Allianz Sports in Barcelona, Spain (July 2022)**

 **Represented India in Commonwealth Chess Championship (2015, 2017& 2018)**